Monthly Performance Report

NovaPay SpA — March 2026

Version 1.0
March 2026
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MONTHLY PERFORMANCE REPORT

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Executive Summary

March was a strong month for NovaPay's performance posture. The P0 fixes from the February Checkup (HikariCP pool + missing index) were deployed in week 1, resulting in a 52% TPS improvement confirmed by the March load test. Uptime exceeded target at 99.96%. Web performance improved significantly — checkout LCP dropped from 4.2s to 2.8s. Two new P2 findings identified, both non-urgent.

Month at a Glance

INDICATOR	VALUE	TARGET	STATUS
Uptime	99.96%	≥99.9%	✅ Pass
p50 Response Time	85ms	<200ms	✅ Pass
p95 Response Time	220ms	<500ms	✅ Pass
Error Rate	0.018%	<1%	✅ Pass
Load Test: Max TPS	488	≥400	✅ Pass
Lighthouse Score (avg)	65/100	≥70	⚠️ Close
RUM LCP (p75)	2.3s	<2.5s	✅ Pass
JS Errors (total)	580	<800	✅ Pass

1. Uptime & Availability

Monthly Uptime Summary

ENDPOINT	UPTIME	AVG RESPONSE	MAX RESPONSE	INCIDENTS
/api/payments	99.98%	75ms	890ms	0
/api/settlements	99.95%	88ms	1,200ms	1
/api/merchants	99.99%	42ms	320ms	0
/checkout	99.93%	105ms	1,800ms	2

Incident Log

DATE	DURATION	ENDPOINT	IMPACT	ROOT CAUSE	RESOLUTION
Mar 5	8 min	/checkout	Slow checkouts	CDN cache invalidation during deploy	Rolling deploy policy updated
Mar 12	12 min	/api/ settlements	Export timeouts	Large merchant batch settlement (120K rows)	Added query limit + pagination
Mar 19	3 min	/checkout	Brief unavailability	Planned deploy v2.14.3	Expected (rolling restart)

Uptime Trend (3 months)

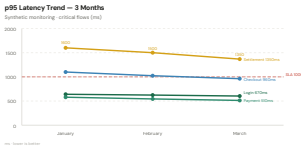
MONTH	UPTIME	INCIDENTS	MTTR
January	99.91%	4	18 min
February	99.92%	3	15 min
March	99.96%	3	8 min

2. Synthetic Monitoring

Critical Flows — Monthly Summary

FLOW	P50	P95	P99	ERROR RATE	CHECKS	STATUS
Login → Dashboard	410ms	670ms	890ms	0.01%	8,640	✅ Pass
Create Payment	305ms	510ms	720ms	0.02%	8,640	✅ Pass
Settlement Export	850ms	1,350ms	1,800ms	0.05%	8,640	✅ Pass
Checkout (guest)	640ms	960ms	1,300ms	0.03%	8,640	✅ Pass

Latency Trend (3 months)



p95 Latency Trend — 3 Months

FLOW	JANUARY P95	FEBRUARY P95	MARCH P95	TREND
Login → Dashboard	720ms	690ms	670ms	Improving
Create Payment	580ms	540ms	510ms	Improving
Settlement Export	1,600ms	1,500ms	1,350ms	Improving
Checkout (guest)	1,100ms	1,020ms	960ms	Improving

3. Load Test Results

Test date: March 15, 2026 **Target:** Validate P0 fix impact, confirm 400+ TPS capacity **Stack:** k6 + Grafana Cloud

Baseline (normal traffic)

METRIC	VALUE	THRESHOLD	STATUS
Virtual Users	50	—	—
Avg TPS	245	≥150	✅ Pass
p50 latency	68ms	<200ms	✅ Pass
p95 latency	180ms	<500ms	✅ Pass
p99 latency	380ms	<1000ms	✅ Pass
Error rate	0.01%	<1%	✅ Pass

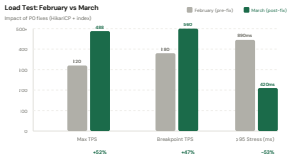
Stress (2x traffic)

METRIC	VALUE	THRESHOLD	STATUS
Virtual Users	100	—	—
Max TPS	488	≥300	✅ Pass
p95 latency	420ms	<1000ms	✅ Pass
Error rate	0.3%	<2%	✅ Pass

Spike (5x burst)

METRIC	VALUE	THRESHOLD	STATUS
Virtual Users	250 (ramp 30s)	—	—
Max TPS	520 (then degradation)	≥400	✅ Pass
p95 latency	1,800ms	<2000ms	✅ Pass
Recovery	18 seconds	<30s	✅ Pass

Load Test Trend (3 months)



Load Test: February vs March

MONTH	MAX TPS	BREAKPOINT	P95 (STRESS)	TREND
January	N/A	N/A	N/A	—
February (Checkup)	320	380	890ms	Baseline
March	488	560	420ms	+52% TPS

> **Impact of P0 fixes confirmed:** Max TPS improved from 320 → 488 (+52%). Breakpoint moved from 380 → 560 TPS. The system now handles 5x spike without exceeding thresholds.

4. Web Performance

Lighthouse CI (weekly runs)

PAGE	LCP	INP	CLS	SCORE	VS LAST MONTH
Homepage	1.9s	78ms	0.04	74/100	+6
Checkout	2.8s	110ms	0.08	58/100	+16
Dashboard	2.6s	88ms	0.06	62/100	+7
API Docs	1.3s	42ms	0.02	91/100	+2

RUM — Real User Monitoring

METRIC	VALUE	TARGET	STATUS
Total Sessions	152,800	—	—
LCP (p75)	2.3s	<2.5s	✅ Pass
INP (p75)	165ms	<200ms	✅ Pass
CLS (p75)	0.07	<0.1	✅ Pass
JS Errors (unique)	12	—	—
JS Errors (total)	580	<800	✅ Pass

Top JS Errors

ERROR	COUNT	PAGE	IMPACT
`TypeError: Cannot read property 'id' of null`	210	/dashboard	Low — edge case with expired sessions
`ResizeObserver loop limit exceeded`	185	/checkout	None — browser noise, can be silenced
`NetworkError: Failed to fetch`	95	/api/*	Low — transient, correlates with poor mobile connections

5. Capacity Planning

SCENARIO	REQUIRED TPS	CURRENT MAX	HEADROOM	STATUS
Current traffic (180 TPS avg)	180	488	171%	✅ Comfortable
2x growth	360	488	36%	✅ OK
5x growth	900	488	-46%	❌ Not ready
10x growth	1,800	488	-73%	❌ Not ready

Capacity Planning

Recommendation

NovaPay is comfortable for current traffic and 2x growth. For 5x+ growth (projected Q4 if partnership deals close), the main bottleneck will be the PostgreSQL connection layer. Recommendations: (1) introduce read replicas for settlement queries, (2) add Redis caching for merchant config (F-03, already planned), (3) consider connection pooler (PgBouncer) for >800 TPS target.

6. Findings & Remediation

New Findings

ID	SEVERITY	FINDING	RECOMMENDED ACTION	ESTIMATED IMPACT
F-07	P2	Settlement export p95 approaching 1.5s for large merchants	Add pagination to batch query (LIMIT 10K per page)	Export latency -60%
F-08	P2	3 noisy JS errors account for 84% of error volume	Silence ResizeObserver, fix null check, add retry for fetch	Cleaner error signal

Previously Reported — Status Update

ID	FINDING	STATUS	NOTES
F-01	HikariCP pool at default (10)	✅ Resolved	Deployed Mar 3 — pool set to 30
F-02	Missing index on settlements	✅ Resolved	Deployed Mar 3 — query from 200ms to 12ms
F-03	No cache on merchant config	In progress	Redis cache PR in review, expected next week
F-04	LCP 4.2s on checkout	In progress	Now 2.8s. JS deferral done, image optimization WIP
F-05	No Redis connection pooling	Open	Scheduled for April sprint
F-06	Health endpoint queries DB	✅ Resolved	Fixed Mar 10 — static 200 response

Remediation Summary

PRIORITY	OPEN	RESOLVED THIS MONTH	TOTAL RESOLVED
P0 Critical	0	2	2
P1 High	2	0	0
P2 Medium	3	1	1

Grafana Dashboard: https://novapay.grafana.net/d/perf-monthly

Loom video walkthrough: https://loom.com/share/novapay-march-2026

XLSX data: attached

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